to-be, 25 new food items have been added to the list, including sugar, oranges, soy and marmite (don’t ask…).

After checking your numbers everyone is in dismay: now you’d have approximately 4 billion different combinations to check.

1.4.3 Algorithms to the Rescue

At that point, it’s clear that you need a computer program to crunch the numbers and take the best decision.

You write one yourself in the next few hours, but even that takes a long time to run, another couple of hours. Now, as much as you’d like to apply the same diet to all the colonists, turns out some of them have allergies: a quarter of them can’t eat gluten, and apparently many swear they are allergic to marmite. So, you’ll have to run the algorithm again a few times, taking each time into account individual allergies. To make thing worse, mission control is considering adding 30 more items to the list, to make up for what people with allergies will miss. If that happens, we would end up with 62 total possible items, and the program will have to go over several billions of billions of possible combinations. You try that, and after a day the program is still running, nowhere close to termination.

The whole team is ready to give up and go back to the potatoes diet, when someone remembers there is a guy in the launch team who had an algorithms book on his desk.